SE 216 – SOFTWARE PROJECT MANAGEMENT
PROJECT RISKS DOCUMENT

PROJECT NAME: Vaccine Integrity Monitoring System (VIMS)

GROUP MEMBERS: Muhammet Ali ÖKAN / Ceren ACAY / Azra AKGÜN / Emir Taha GÖKÇE / Ömer Talha ÇEKMEZ / Kayra Alp UŞAKLI / Hakkı Can DURAN / Ege ENGİN

LIKELIHOOD RANK	RISK DESCRIPTION
1	**Technical Skill Gap Risk** - The team has little experience with IoT data streams and real-time AI predictive models, which may slow down initial development.
2	**Hardware Integration Failure -** Integrating the software with external smart storage cases and IoT temperature sensors might fail or require more time than planned.
3	**Inadequate Testing Environment -** It will be difficult to test the system realistically without access to an actual logistics network or constant physical network drops.
4	**AI Training Data Scarcity -** Finding an accurate historical dataset of temperature fluctuations to train the AI predictive model might be difficult.
5	**Key Personnel Turnover -** A critical team member drops out or becomes unavailable, leaving the rest of the team struggling to cover the missing role.
6	**Scope Creep -** Scope creep regarding what specific details must be included in the QR-based Arrival Report might change as development progresses.
7	**Server Performance Bottleneck -** The local development servers might struggle to process and display sensor data arriving every 60 seconds without lagging.
8	**Poor Team Coordination -** Poor coordination among team members could lead to overlapping work or missed deadlines for sprints.
9	**Security & Data Privacy Vulnerability -** Sensitive shipment data could be exposed due to improper TLS implementation or weak authentication.
10	**Regulatory Compliance Failure -** The system may fail to meet international regulatory and certification standards, blocking real-world deployment.

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IMPACT RANK	RISK DESCRIPTION
1	**Key Personnel Turnover -** A critical team member drops out or becomes unavailable, leaving the rest of the team struggling to cover the missing role.
2	**Hardware Integration Failure -** Integrating the software with external smart storage cases and IoT temperature sensors might fail or require more time than planned.
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LIKELIHOOD RANK	IMPACT RANK	COMBINED RANK	RISK DESCRIPTION
2	2	4	**Hardware Integration Failure -** Integrating the software with external smart storage cases and IoT temperature sensors might fail or require more time than planned.
5	1	6	**Key Personnel Turnover -** A critical team member drops out or becomes unavailable, leaving the rest of the team struggling to cover the missing role.
1	6	7	**Technical Skill Gap Risk** - The team has little experience with IoT data streams and real-time AI predictive models, which may slow down initial development.
4	3	7	**AI Training Data Scarcity -** Finding an accurate historical dataset of temperature fluctuations to train the AI predictive model might be difficult.
3	4	7	**Inadequate Testing Environment -** It will be difficult to test the system realistically without access to an actual logistics network or constant physical network drops.
10	5	15	**Regulatory Compliance Failure -** The system may fail to meet international regulatory and certification standards, blocking real-world deployment.
7	8	15	**Server Performance Bottleneck -** The local development servers might struggle to process and display sensor data arriving every 60 seconds without lagging.
6	9	15	**Scope Creep -** Scope creep regarding what specific details must be included in the QR-based Arrival Report might change as development progresses.
9	7	16	**Security & Data Privacy Vulnerability -** Sensitive shipment data could be exposed due to improper TLS implementation or weak authentication.
8	10	18	**Poor Team Coordination -** Poor coordination among team members could lead to overlapping work or missed deadlines for sprints.

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